

HR Analytics — Employee Attrition Analysis

Business Problem & Analytical Objective

1. Background

Employee attrition is an important workforce management challenge because the loss of employees can create additional recruitment costs, increase workload for existing employees, disrupt team productivity, and result in the loss of organizational knowledge and experience.

For HR management, understanding **who is leaving, where attrition is concentrated, and which employee or workplace characteristics are associated with higher turnover** is essential for developing effective employee retention strategies.

This project uses employee-level HR data to investigate workforce composition and identify patterns associated with employee attrition.

2. Business Problem

The organization has **1,470 employees**, with **237 employees recorded as having left the organization**, resulting in an overall attrition rate of **16.12%**.

However, the overall attrition rate alone does not explain the underlying workforce problem.

The key business challenge is to determine whether employee turnover is concentrated among particular:

- Age groups
- Gender groups
- Departments
- Job roles
- Job levels
- Salary bands
- Tenure groups
- Satisfaction levels
- Overtime categories
- Business travel categories
- Stock option levels

Without this segmentation, HR management may find it difficult to identify the employees and organizational areas requiring targeted retention interventions.

Therefore, this analysis focuses on transforming employee data into **actionable workforce intelligence** that can support evidence-based HR decision-making.

3. Core Business Question

The central business question guiding this project is:

What employee, compensation, workplace, and career-related factors are associated with higher employee attrition, and which workforce segments should HR prioritize for retention efforts?

4. Analytical Objectives

The analysis was designed to achieve the following objectives:

Objective 1 — Measure Overall Attrition

Determine:

- Total number of employees
 - Number of employees who have left
 - Number of active employees
 - Overall employee attrition rate
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Objective 2 — Understand Workforce Demographics

Analyze employee distribution by:

- Gender
- Age group
- Marital status
- Education level
- Education field

This provides context for understanding the composition of the workforce.

Objective 3 — Analyze Organizational Structure

Examine employee distribution across:

- Departments
- Job roles
- Job levels

The objective is to determine where workforce concentration and potential attrition risks exist within the organization.

Objective 4 — Investigate Compensation

Analyze:

- Average monthly income
- Salary bands
- Average salary by department
- Average salary by job role
- Average salary by job level

The purpose is to determine whether compensation patterns are associated with differences in employee attrition.

Objective 5 — Investigate Employee Satisfaction

Evaluate whether attrition varies across different levels of:

- Job satisfaction
- Environment satisfaction
- Work-life balance

This helps assess whether employee experience is associated with retention outcomes.

Objective 6 — Examine Workplace Engagement

Investigate the relationship between employee attrition and:

- Overtime
- Business travel

The objective is to identify whether workload and travel requirements correspond with higher employee turnover.

Objective 7 — Analyze Career Progression

Evaluate attrition across:

- Job levels
- Tenure groups
- Stock option levels

This helps identify whether employees at particular stages of their careers experience greater turnover.

Objective 8 — Identify High-Risk Employee Segments

Rank employee groups according to their attrition rates to identify segments requiring further HR investigation.

The analysis specifically seeks to answer questions such as:

Which department has the highest attrition?

Which job role has the highest attrition?

Which age group is most affected?

Which salary band has the highest attrition?

Does overtime correspond with higher employee turnover?

Does job satisfaction vary with attrition?

Which tenure group represents the greatest retention risk?

5. Key Analytical Questions

The analysis was structured around **26 core business questions**, divided into two major analytical stages.

Stage 1 — Workforce & Employee Overview

The first stage establishes the organization's workforce profile by examining:

1. How many employees does the company have?
2. How many employees have left?
3. How many employees remain active?
4. What is the overall attrition rate?
5. What is the average employee age?

6. What is the average monthly income?
 7. How is the workforce distributed by gender?
 8. How is the workforce distributed by age group?
 9. What is the marital-status distribution?
 10. What is the education-level distribution?
 11. What are the major education fields?
 12. How are employees distributed across departments?
 13. How are employees distributed across job roles?
 14. How are employees distributed across job levels?
 15. Which departments have the highest average salary?
 16. What is the overtime distribution?
 17. How frequently do employees travel for business?
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6. Stage 2 — Attrition Analysis

The second stage focuses specifically on identifying employee turnover patterns.

The analysis investigates:

1. Which gender has the highest attrition rate?
 2. Which age group experiences the highest attrition?
 3. Which department loses the most employees?
 4. Which job role has the highest attrition?
 5. Which job level experiences the highest turnover?
 6. Does salary level influence attrition?
 7. Does job satisfaction affect attrition?
 8. Does environment satisfaction affect attrition?
 9. Does work-life balance affect attrition?
 10. Does overtime increase attrition?
 11. Does business travel affect attrition?
 12. Which tenure group has the highest attrition?
 13. Does stock option level influence attrition?
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7. Advanced Analytical Objectives

Beyond basic descriptive analysis, advanced SQL techniques were applied to investigate deeper workforce patterns.

The advanced analysis examined:

- Department salary rankings
- Highest-paid employees
- Employee salary compared with departmental averages
- Employees earning above their department average

- Department attrition rankings
- Running total of monthly income
- Average salary by job level
- Workforce percentage by department
- Department attrition compared with the company average
- Top five highest-paid employees within each department

These analyses demonstrate how SQL can be used not only for data retrieval but also for **comparative analysis, ranking, segmentation, and business intelligence**.

8. Analytical Approach

The project followed a structured analytical workflow:

Step 1 — Data Acquisition

The original **IBM HR Analytics Employee Attrition** dataset was obtained and imported into MySQL.

Step 2 — Data Validation

The dataset was checked for:

- Row count
- Column structure
- Primary-key integrity
- Missing values
- Duplicate employee records
- Valid categorical values
- Employee identifier consistency

The final validated dataset contained:

1,470 employee records and 35 variables.

Step 3 — Exploratory Data Analysis

SQL was used to understand workforce demographics, organizational structure, compensation, employee satisfaction, and workplace characteristics.

Step 4 — Attrition Analysis

Attrition rates were calculated across multiple employee and organizational dimensions to identify high-risk segments.

Step 5 — Advanced SQL Analysis

CTEs, window functions, ranking functions, conditional logic, aggregations, and comparative calculations were used to answer more complex business questions.

Step 6 — Power BI Visualization

The analytical results were transformed into a five-page interactive Power BI dashboard consisting of:

1. **Executive Overview**
2. **Attrition Analysis**
3. **Workforce Insights**
4. **Compensation Analysis**
5. **HR Insights & Recommendations**

Step 7 — Business Interpretation

The analytical findings were translated into practical HR insights and recommendations focused on employee retention.

9. Key Performance Indicators

The project uses several core HR metrics to evaluate workforce health.

Workforce KPIs

- **Total Employees:** 1,470
- **Active Employees:** 1,233
- **Employees Who Left:** 237
- **Overall Attrition Rate:** 16.12%
- **Average Employee Age:** 36.9
- **Average Monthly Income:** 6,502.93

Attrition KPI

The primary attrition metric is:

$$\text{Attrition Rate} = \text{Employees Who Left} \div \text{Total Employees} \times 100$$

This metric is used consistently throughout the analysis to enable meaningful comparison across employee segments.

10. Business Intelligence Perspective

The purpose of this project is not simply to calculate attrition rates.

The analysis aims to answer the broader business question:

Where should HR management focus its attention to reduce employee turnover?

For example, identifying an overall attrition rate of **16.12%** provides useful context, but identifying that **Sales Representatives have a 39.76% attrition rate** provides a much more actionable signal.

Similarly, identifying that employees working overtime have a **30.53% attrition rate** allows management to investigate workload and employee well-being more specifically.

This approach moves the analysis from **descriptive reporting** toward **diagnostic workforce intelligence**.

11. Expected Business Value

The analysis is intended to help HR management:

- Identify high-risk employee segments
- Prioritize retention initiatives
- Understand workforce composition
- Monitor departmental turnover
- Evaluate compensation patterns
- Identify potential workload-related risks
- Understand employee satisfaction patterns
- Improve workforce planning
- Support evidence-based HR decision-making

The resulting Power BI dashboard provides management with an interactive platform for monitoring these workforce indicators.

12. Scope and Analytical Limitation

This analysis identifies **patterns and associations** between employee characteristics and attrition.

The results should not automatically be interpreted as evidence that a particular factor directly causes an employee to leave.

For example, the higher attrition rate among overtime employees indicates an association between overtime status and attrition within this dataset. It does not, by itself, establish that overtime is the direct cause of employee turnover.

Further investigation using employee surveys, qualitative feedback, longitudinal data, compensation benchmarks, and additional organizational variables would be required to establish causal relationships.

13. Analytical Outcome

The final outcome of this project is a complete HR analytics solution that combines:

Raw Employee Data → Data Validation → SQL Analysis → Advanced Analytics → Power BI Visualization → Business Insights → HR Recommendations

The project demonstrates how structured data analysis can support HR management in moving from "**How many employees are leaving?**" to the more valuable questions of:

"Who is most at risk?"

"Where is attrition concentrated?"

"What workforce characteristics are associated with higher turnover?"

"Where should management prioritize retention efforts?"

Tools Used

MySQL | SQL | Microsoft Power BI | DAX | Excel

Project Deliverables

- **SQL Analysis** — Data validation, EDA, attrition analysis, and advanced SQL.
- **Power BI Dashboard** — Five-page interactive HR analytics dashboard.
- **Executive Summary** — Key findings and business significance.
- **Business Problem & Analytical Objective** — Business context, analytical questions, and methodology.
- **Insights & Recommendations** — Detailed findings and recommended HR actions.